

Gibbs Transfer Energies of Ions from a Mixed Solvent of 2H,3H-Decafluoropentane and 1,2-Dichloroethane to Water

Hajime KATANO,*† Kohei UEMATSU,* Yasuhiro KURODA,* and Toshiyuki OSAKAI**

*Department of Bioscience and Biotechnology, Fukui Prefectural University, Eiheiiji, Fukui 910-1195, Japan

**Department of Chemistry, Kobe University, Nada, Kobe 657-8501, Japan

The transfer of hydrophilic, lipophilic, and fluorophilic ions at the interface between water (W) and a mixed solvent (MIX) of 2H,3H-decafluoropentane (DFP) and 1,2-dichloroethane (DCE) was studied voltammetrically and potentiometrically, and the formal Gibbs transfer energies of the ions from MIX to W, $\Delta G_{\text{tr,MIX} \rightarrow \text{W}}^0$, were determined. The $\Delta G_{\text{tr,MIX} \rightarrow \text{W}}^0$ values of all the ions tested were higher than those from DFP to W. Namely, the ions would exist more stably in MIX than DFP, even for fluorophilic ions. This is due to the addition of DCE, which has a higher dielectric constant. A comparison of $\Delta G_{\text{tr,MIX} \rightarrow \text{W}}^0$ with that from DCE to W showed a superior affinity of fluorophilic ions to the fluoruous solvent in spite of equivolume addition of DCE. Therefore, the mixed solvent would be a practically superior extraction medium for fluorophilic ions. In practice, the MIX | methylene blue⁺ (W) system showed higher extractability of a fluorophilic ion $\text{C}_8\text{F}_{17}\text{SO}_3^-$ than the DFP | W and DCE | W systems.

Keywords Gibbs transfer energy, 2H,3H-decafluoropentane, 1,2-dichloroethane, mixed solvent, fluorophilic ion

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Introduction

In previous papers,^{1,2} we reported on the applicability of a fluoruous solvent, 2H,3H-decafluoropentane (DFP), as a non-aqueous solvent for liquid-liquid electrochemistry, and voltammetric and potentiometric observation for the transfer of various monovalent ions at the DFP | water (W) interface. From the electrochemical data, the formal Gibbs transfer energies of the ions from DFP to W, $\Delta G_{\text{tr,DFP} \rightarrow \text{W}}^0$, were determined. The $\Delta G_{\text{tr,DFP} \rightarrow \text{W}}^0$ values were compared with those from an organic solvent, 1,2-dichloroethane (DCE), to W, $\Delta G_{\text{tr,DCE} \rightarrow \text{W}}^0$, as a quantitative scale for hydrophilicity/hydrophobicity of ions. By introducing $\Delta G_{\text{tr,DCE} \rightarrow \text{DFP}}^0$ ($= \Delta G_{\text{tr,DFP} \rightarrow \text{W}}^0 - \Delta G_{\text{tr,DCE} \rightarrow \text{W}}^0$) as a quantitative scale for fluorophilicity/lipophilicity, the ions tested could be divided into three categories: hydrophilic ions with $\Delta G_{\text{tr,DCE} \rightarrow \text{W}}^0 < 0$ and $\Delta G_{\text{tr,DCE} \rightarrow \text{DFP}}^0 \approx 0$, lipophilic ions with $\Delta G_{\text{tr,DCE} \rightarrow \text{W}}^0 > 0$ and $\Delta G_{\text{tr,DCE} \rightarrow \text{DFP}}^0 > 0$, and fluorophilic ions with $\Delta G_{\text{tr,DCE} \rightarrow \text{W}}^0 > 0$ and $\Delta G_{\text{tr,DCE} \rightarrow \text{DFP}}^0 < 0$. The bulky and highly-fluorinated ions, such as PF_6^- and bis(perfluoroalkanesulfonyl)-imide ($\text{C}_n\text{F}_{2n+1}\text{SO}_2)_2\text{N}^-$ with $n = 1 - 3$, correspond to the fluorophilic ions.

The above results suggest that DFP is an essentially suitable extraction medium for fluorophilic ions. However, the authors have found no easily available fluorophilic cationic dye, although the preparation of some fluorophilic cations have been reported.³⁻⁶ Accordingly, a lipophilic cationic dye should be used in an extraction-spectrophotometric determination, and then a mixed solvent of DFP and DCE may become a superior extraction medium. In this study, the electrochemical study was

extended to the interfacial system between a 1:1 (v/v) DFP-DCE mixed solvent (MIX) and W. The transfer energies from MIX to W, $\Delta G_{\text{tr,MIX} \rightarrow \text{W}}^0$, were determined, and were compared with $\Delta G_{\text{tr,DFP} \rightarrow \text{W}}^0$ and $\Delta G_{\text{tr,DCE} \rightarrow \text{W}}^0$. A prediction using the thus determined transfer energies was successful for the solvent extraction of a fluorophilic anion, perfluorooctane sulfonate ($\text{C}_8\text{F}_{17}\text{SO}_3^-$), with a lipophilic cationic dye, methylene blue (MB^+).

Experimental

Chemicals

DFP and DCE were purified as previously mentioned.^{1,7} The tetra-*n*-octylammonium (Oct_4N^+) salt of bis(nonafluorobutanesulfonyl)imide (BNSI^-), the tetraphenylarsonium (Ph_4As^+) salt of BNSI^- , the tetrabutylammonium (Bu_4N^+) salts of tetrakis[3,5-bis(trifluoromethyl)phenyl]borate (TFPB^-) and ($\text{C}_n\text{F}_{2n+1}\text{SO}_2)_2\text{N}^-$ with $n = 1 - 3$ were prepared as previously described.^{1,2} The Bu_4N^+ salts with perfluorooctanoate ($\text{C}_7\text{F}_{15}\text{COO}^-$) and $\text{C}_8\text{F}_{17}\text{SO}_3^-$ anions were prepared in a manner similar to that used for the preparation of $\text{Bu}_4\text{N}(\text{C}_n\text{F}_{2n+1}\text{SO}_2)_2\text{N}$. Tetrabutylammonium chloride (Bu_4NCl) was obtained from FUJIFILM Wako Pure Chemical Corp., and was purified by metathesis with silver chloride.⁸ Other chemicals were of analytical grade and used as received.

Measurements

Ion-transfer reactions across the MIX | W interface have been studied by means of cyclic voltammetry using a three-electrode system. Cells-I and -II represent the electrochemical cells used for recording cyclic voltammograms (CVs) of the transfer of Ph_4As^+ and tetraphenylborate ion (Ph_4B^-) across the MIX | W interface, respectively.

† To whom correspondence should be addressed.

E-mail: hajime@fpu.ac.jp

cell-I:

	Phase I	Phase II	Phase III	Phase IV	
Pt	<i>a</i> mM Ph ₄ AsBNSI 0.1 M Oct ₄ NBNSI	<i>a</i> mM Ph ₄ AsBNSI 0.1 M Oct ₄ NBNSI	0.05 M MgSO ₄	0.1 M LiCl	AgCl Ag
(CE)	(MIX)	(MIX)	(W)	(W)	(RE1)
AgCl Ag	0.1 M LiCl 0.1 M 0.1 M LiOH HBNSI	0.1 M Oct ₄ NBNSI			
(RE2)	(W)	(MIX)			

cell-II:

	Phase I	Phase II	Phase III	Phase IV	
Pt	0.1 M Bu ₄ NBNSI	0.1 M Bu ₄ NTPPB	<i>b</i> mM NaPh ₄ B 0.1 M LiCl	0.1 M LiCl	AgCl Ag
(CE)	(MIX)	(MIX)	(W)	(W)	(RE1)
AgCl Ag	0.1 M Bu ₄ NCl	0.1 M Bu ₄ NTPPB			
(RE2)	(W)	(MIX)			

The test MIX | W interface is indicated by an asterisk. The electrolysis cell is described in Ref. 1. The geometric area of the test interface was 0.126 cm². The applied potential, E , is defined as the terminal potential of RE1 referred to that of RE2. The solution resistance between the two reference electrodes was about 2 k Ω for cells-I and -II. E was controlled by a potentiostat furnished with a positive feedback iR compensation circuit.

The formal potential for the transfer of j^- ($=(\text{C}_2\text{F}_5\text{SO}_2)_2\text{N}^-$ and $(\text{C}_3\text{F}_7\text{SO}_2)_2\text{N}^-$) at the MIX | W interface was determined by the EMF measurement with cell-III.

cell-III:

	Phase I	Phase II	Phase III	Phase IV	
Ag AgCl	0.1 M Bu ₄ NCl	0.1 M Bu ₄ N j^-	<i>c</i> mM Li j^- or H j^- 0.1 M LiCl	0.1 M LiCl	AgCl Ag
(RE2)	(W)	(MIX)	(W)	(W)	(RE1)

The EMF was measured by a digital multimeter (Yokogawa 7542).

The dielectric constant of MIX was determined by a liquid dielectric constant meter (Nihon Rufuto Model 871). The viscosity of MIX was determined by a vibro viscometer (A & D SV-10). All of the measurements were carried out at 25 $\pm$ 1 $^\circ$ C.

Results and Discussion

The dielectric constant of MIX was determined to be 8.3 at 25 $^\circ$ C. This value is higher than 6.72 of DFP,² and is lower than 10.45 of DCE.³ MIX as well as DFP did not dissolve the Bu₄N⁺ and Oct₄N⁺ salts of lipophilic Ph₄B⁻ anion at 1 mM level. However, MIX dissolved the Bu₄N⁺ and Oct₄N⁺ salts of highly-fluorinated anions, BNSI⁻, TFPB⁻, and C₈F₁₇SO₃⁻, at 0.1 M level, suggesting a fluorous character of the mixed solvent.

Because of the lower dielectric constant of the solvents tested, ion-pairing between the electrolyte cations and anions may occur in the solutions. The association constants of the ion-pairs (Bu₄N⁺-BNSI⁻, Bu₄N⁺-TFPB⁻, Bu₄N⁺-C₈F₁₇SO₃⁻, and Oct₄N⁺-BNSI⁻) in α = MIX, DFP, and DCE, K_a^α , were

Table 1 Association constants of the ion-pairs in α (= DFP, DCE, DFP, and DEC), K_a^α

Ion-pair	MIX	DFP	DCE
Bu ₄ N ⁺ -BNSI ⁻	1.7×10^4	1.5×10^5	$4.6 \times 10_3$
Bu ₄ N ⁺ -TFPB ⁻	2.6×10^3	2.9×10^4	9.2×10^2
Bu ₄ N ⁺ -C ₈ F ₁₇ SO ₃ ⁻	1.7×10^4	1.8×10^5	7.5×10^3
Oct ₄ N ⁺ -BNSI ⁻	1.5×10^4	1.4×10^5	4.2×10^3

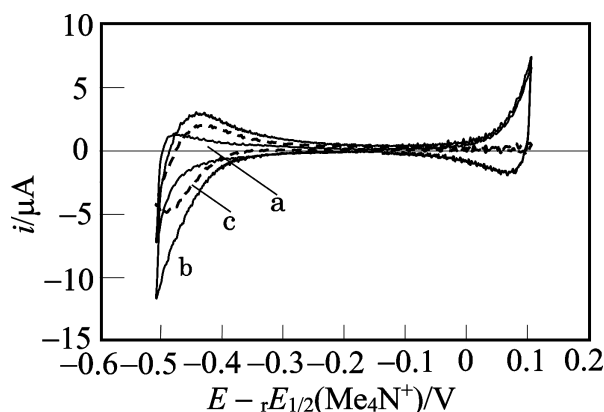


Fig. 1 Cyclic voltammograms at the 0.1 M Oct₄NBNSI (MIX) | 0.05 M MgSO₄ (W) interface in the (a) absence and (b) presence of 0.5 mM Ph₄AsBNSI in the MIX phase. Scan rate, ν = 0.1 V s⁻¹. The dashed line (c) represents the difference, (b) - (a).

determined by conductivity,⁹ as listed in Table 1. The K_a^{MIX} values for each electrolyte are about 10-fold smaller than K_a^{DFP} , indicating that the chemical potentials of the ionic species in MIX are lower than those in DFP.

In Fig. 1, curve (a) shows a CV obtained with cell-I at $a = 0$, that is, the base current of the 0.1 M Oct₄NBNSI (MIX) | 0.05 M MgSO₄ (W) interface at the scan rate of $\nu = 0.1$ V s⁻¹. A polarizable potential window was observed. In the figure, the E value is reported against the reversible half-wave potential of the transfer of the tetramethylammonium (Me₄N⁺) ion across the MIX | W interface, which was determined to be $E_{1/2}(\text{Me}_4\text{N}^+) = -0.212 \pm 0.002$ V with cell-I, in which Phase III was replaced by 0.50 mM Me₄NBr and 0.1 M LiCl (W) and $a = 0$.

Curve (b) in Fig. 1 shows a CV obtained with cell-I at $a = 0.5$ and $\nu = 0.1$ V s⁻¹. After the correction for the base current (curve (c)), a pair of well-defined cathodic and anodic peak currents were observed, corresponding to the transfer of the Ph₄As⁺ ion from the MIX to the W phase and back to the MIX phase, respectively. The height of the cathodic peak current, corrected for the baseline, was proportional to $\nu^{1/2}$. The potential peak separation increased slightly with $\nu^{1/2}$. This could be ascribed to the iR potential drop due to the uncompensated solution resistance. The extrapolation of the peak separation vs. $\nu^{1/2}$ plot to $\nu^{1/2} = 0$ gave 0.056 ± 0.005 V. The midpoint potential remained independently of ν . The results indicate that the voltammogram can be assigned to a reversible CV of monovalent ion-transfer.¹⁰ By taking that the reversible half-wave potential is equal to the midpoint potential, $E_{1/2}(\text{Ph}_4\text{As}^+)$ was determined to be -0.453 ± 0.003 V vs. $E_{1/2}(\text{Me}_4\text{N}^+)$.

The CVs of the transfer of the Ph₄B⁻ ion at the MIX | W interface were recorded with cell-II. As shown in Fig. 2, Ph₄B⁻ also gave a well-developed voltammetric wave within the

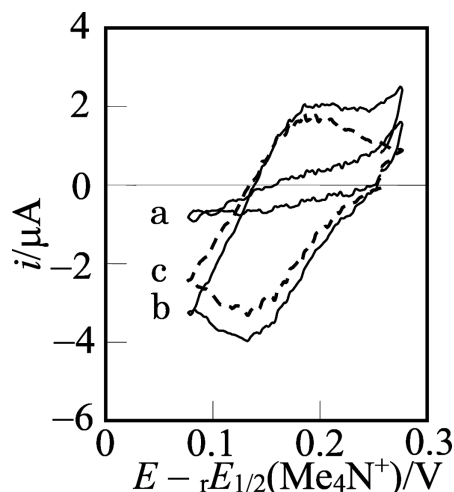


Fig. 2 CVs at the 0.1 M Bu₄NTPFB (MIX) | 0.1 M LiCl (W) interface in the (a) absence and (b) presence of 0.1 mM NaPh₄B in the W phase. $\nu = 0.1 \text{ V s}^{-1}$. The dashed line (c) represents the difference, (b) – (a).

potential window. The dependence of the peak currents and peak potentials on $\nu^{1/2}$ was the same as that of Ph₄As⁺, indicating that the voltammetric behaviors of these ions are of reversible nature. The reversible half-wave potential was determined as $rE_{1/2}(\text{Ph}_4\text{B}^-) = 0.160 \pm 0.004 \text{ V vs. } rE_{1/2}(\text{Me}_4\text{N}^+)$.

The reversible half-wave potential for the transfer of a j^z -ion can be related to the formal potential in terms of the Galvani potential difference at the MIX | W interface, $\Delta_{\text{MIX}}^W\phi^{0'}$, as given by

$$rE_{1/2} = \Delta_{\text{MIX}}^W\phi^{0'} + (RT/zF)\ln[(D^W/D^{\text{MIX}})^{1/2}] + \Delta E_{\text{ref}}, \quad (1)$$

where ΔE_{ref} is determined by the reference electrode system for cells-I and -II, and D^α ($\alpha = \text{W}$ and MIX) the diffusion coefficient of j^z -ion in α phase. R , T , and F are used in their usual meanings. The viscosity of MIX was determined to be 0.87 cP at 25°C, which is close to that of W. On the basis of the Walden rule,¹¹ the ratio of the diffusion is assumed to be $D^W/D^{\text{MIX}} = 1$, that is, $(RT/zF)\ln[(D^W/D^{\text{MIX}})^{1/2}] = 0$. Based on the so-called “TATB” assumption,¹² we assume

$$\Delta_{\text{MIX}}^W\phi^{0'}(\text{Ph}_4\text{As}^+) + \Delta_{\text{MIX}}^W\phi^{0'}(\text{Ph}_4\text{B}^-) = 0. \quad (2)$$

Thus, $\Delta_{\text{MIX}}^W\phi^{0'}$ values are calculated to be -0.307 ± 0.003 , 0.307 ± 0.004 , and $0.147 \pm 0.004 \text{ V}$ for Ph₄As⁺, Ph₄B[−], and Me₄N⁺, respectively.

The CVs of the transfer of $j^+ = \text{choline}^+$, acetylcholine⁺, MB⁺, tetraethylammonium (Et₄N⁺), tetrapropylammonium (Pr₄N⁺), and Bu₄N⁺ at the MIX | W interface were recorded with cell-I, in which Phase III (W) was replaced by 0.50 mM $j^+\text{Cl}^-$ or $j^+\text{Br}^-$ and 0.05 M MgSO₄ (W), and Phases I and II were replaced by a MIX solution containing 0.1 M Oct₄NBNSI only. The CVs of the transfer of $j^- = 2,4,6\text{-trinitrophenolate}$ (TNP[−]), 2,4-dinitrophenolate (DNP[−]), ClO₄[−], SCN[−], I[−], NO₃[−], and Br[−] at the MIX | W interface were recorded with cell-I, in which Phase III (W) was replaced by 0.50 mM Na⁺ j^- and 0.05 M MgSO₄ (W) and Phases I and II were replaced by 0.1 M Oct₄NBNSI (DFP). These ions gave a reversible voltammetric wave within the potential window, and their $\Delta_{\text{MIX}}^W\phi^{0'}$ values were determined, as listed in Table 2. In the table, the formal potentials of their ion-transfer at the DFP | W and DCE | W interfaces^{2,7} are also listed.

Table 2 Formal ion-transfer potentials (in V) of non-fluorinated ions at the MIX | W, DFP | W, and DCE | W interfaces

Ion	$\Delta_{\text{MIX}}^W\phi^{0'}$	$\Delta_{\text{DFP}}^W\phi^{0'a}$	$\Delta_{\text{DCE}}^W\phi^{0'b}$
Cation			
Choline ⁺	0.206 ± 0.002	0.223	0.200
Me ₄ N ⁺	0.147 ± 0.004	0.160	0.133
Acetylcholine ⁺	0.121 ± 0.003	0.140	0.110
Et ₄ N ⁺	0.023 ± 0.002	0.036	−0.013
Pr ₄ N ⁺	-0.108 ± 0.003	−0.087	−0.141
MB ⁺	-0.168 ± 0.002	−0.084	−0.223
Bu ₄ N ⁺	-0.224 ± 0.003	−0.189	−0.278
Ph ₄ As ⁺	-0.307 ± 0.003	−0.227	−0.341
Anion			
Ph ₄ B [−]	0.307 ± 0.004	0.227	0.345
TNP [−]	0.061 ± 0.003	−0.016	0.043
ClO ₄ [−]	-0.061 ± 0.003	−0.097	−0.078
DNP [−]	-0.064 ± 0.002	−0.113	−0.083
SCN [−]	-0.155 ± 0.002	−0.197	−0.161
I [−]	-0.181 ± 0.002	−0.227	−0.179
NO ₃ [−]	-0.219 ± 0.002	−0.241	−0.258
Br [−]	-0.281 ± 0.003	−0.318	−0.289

a. Ref. 2.

b. Ref. 7.

Table 3 Formal ion-transfer potentials (in V) of highly-fluorinated ions at the MIX | W, DFP | W, and DCE | W interfaces

Ion	$\Delta_{\text{MIX}}^W\phi^{0'}$	$\Delta_{\text{DFP}}^W\phi^{0'c}$	$\Delta_{\text{DCE}}^W\phi^{0'd}$
(C ₃ F ₇ SO ₂) ₂ N [−]	0.320 ± 0.003^a	0.317	0.206
(C ₂ F ₅ SO ₂) ₂ N [−]	0.232 ± 0.002^a	0.217	0.142
C ₈ F ₁₇ SO ₃ [−]	0.211 ± 0.004^a	0.176	0.119 ^a
(CF ₃ SO ₂) ₂ N [−]	0.155 ± 0.004^a	0.132	0.078
C ₇ F ₁₅ COO [−]	0.092 ± 0.003^a	0.091	0.047 ^a
PF ₆ [−]	0.064 ± 0.003^b	0.063	0.041
C ₄ F ₉ COO [−]	-0.079 ± 0.003^b	−0.099	−0.103
BF ₄ [−]	-0.093 ± 0.002^b	−0.113	−0.097
C ₃ F ₇ COO [−]	-0.129 ± 0.004^b	−0.156	−0.140
C ₂ F ₅ COO [−]	-0.177 ± 0.004^b	−0.199	−0.181
CF ₃ COO [−]	-0.221 ± 0.002^b	−0.264	— ^c

a. With cell-III.

b. With cell-I.

c. Ref. 2.

d. Ref. 7.

e. Out of the potential window.

Highly-fluorinated anions, BF₄[−], PF₆[−], C_nF_{2n+1}COO[−] ($n = 1, 2, 3$, and 4), and (CF₃SO₂)₂N[−] gave reversible CVs within the potential window at the 0.1 M Oct₄NBNSI (MIX) | 0.05 M MgSO₄ (W) interface, and their $\Delta_{\text{MIX}}^W\phi^{0'}$ values were determined, as listed in Table 3. The $\Delta_{\text{MIX}}^W\phi^{0'}$ values for (C₂F₅SO₂)₂N[−] and (C₃F₇SO₂)₂N[−] were determined from the EMF measurement with cell-III. Plots of EMF vs. $\log(c_j^W/c_j^{\text{MIX}})$ for $j^- = (\text{C}_2\text{F}_5\text{SO}_2)_2\text{N}^-$ and $(\text{C}_3\text{F}_7\text{SO}_2)_2\text{N}^-$ with c_j^α being the concentration of j^- ion in α phase ($\alpha = \text{MIX}$ and W), gave regression lines with the slope of 0.056 ± 0.004 and $0.059 \pm 0.007 \text{ V/dec}$, respectively. The values consist with the theoretical one for the Nernst equation for monovalent charge transfer within the experimental errors. From the intercept, $\Delta_{\text{MIX}}^W\phi^{0'}$ values were determined to be 0.230 ± 0.002 and $0.320 \pm 0.003 \text{ V}$ for (C₂F₅SO₂)₂N[−] and (C₃F₇SO₂)₂N[−], respectively. Similarly, for $j^- = \text{C}_7\text{F}_{15}\text{COO}^-$ and C₈F₁₇SO₃[−], the EMF measurement was performed with cell-III, in which Phase II was replaced by the 0.1 M Bu₄N⁺ j^- .

Table 4 Formal Gibbs energy (in kJ mol⁻¹) for the transfer of tested ions from MIX, DFP, and DCE to W

Ion	$\Delta G_{tr,MIX \rightarrow W}^{0'}$	$\Delta G_{tr,DFP \rightarrow W}^{0'}$	$\Delta G_{tr,DCE \rightarrow W}^{0'}$
Cation			
Choline ⁺	-19.9	-21.5	-19.3
Me ₄ N ⁺	-14.2	-15.4	-12.8
Acetylcholine ⁺	-11.6	-13.5	-10.6
Et ₄ N ⁺	-2.2	-3.5	1.3
Pr ₄ N ⁺	10.4	8.4	16.2
MB ⁺	16.2	8.1	21.5
Bu ₄ N ⁺	21.9	18.2	26.8
Ph ₄ As ⁺	29.6	21.9	32.9
Anion			
Ph ₄ B ⁻	29.6	21.9	33.3
TNP ⁻	5.9	-1.5	4.1
ClO ₄ ⁻	-5.9	-9.4	-7.5
DNP ⁻	-6.2	-10.9	-8.0
SCN ⁻	-14.9	-19.0	-15.5
I ⁻	-17.4	-21.9	-17.3
NO ₃ ⁻	-21.2	-23.3	-24.9
Br ⁻	-27.1	-30.7	-27.9
Highly-fluorinated			
(C ₃ F ₇ SO ₂) ₂ N ⁻	30.9	30.6	19.9
(C ₂ F ₅ SO ₂) ₂ N ⁻	22.4	20.9	13.7
C ₈ F ₁₇ SO ₃ ⁻	21.3	17.0	11.5
(CF ₃ SO ₂) ₂ N ⁻	15.0	12.9	7.5
C ₇ F ₁₅ COO ⁻	8.8	8.8	4.5
PF ₆ ⁻	6.2	6.1	4.0
C ₄ F ₉ COO ⁻	-7.6	-9.6	-9.9
BF ₄ ⁻	-9.0	-10.9	-9.4
C ₃ F ₇ COO ⁻	-12.5	-15.1	-13.5
C ₂ F ₅ COO ⁻	-17.1	-19.2	-17.5

(MIX, DFP, or DCE). The EMF showed Nernstian behaviors. The resulting $\Delta_a^w \phi^{0'}$ values are also listed in Table 3.

Because of the large K_a^α values (α = MIX, DFP, and MIX), the experimental $\Delta_a^w \phi^{0'}$ is an apparent value, as written by¹³

$$\Delta_a^w \phi^{0'} = \Delta_a^w \phi^0 + (RT/zF) \ln(\gamma^\alpha/\gamma^w) - (RT/zF) \ln(1 + K_a^\alpha \chi c_{SE}^\alpha), \quad (3)$$

where $\Delta_a^w \phi^0$ is the standard potential for the transfer of the j^z -ion at the α | W interface, γ^α the activity coefficient of j^z -ion in α phase, χ the degree of dissociation of the supporting electrolyte, and c_{SE}^α the bulk concentration of the supporting electrolyte in the α phase. In the derivation of Eq. (3), for simplicity, the diffusion coefficient of the ion-pair in the α phase is assumed to be equal to those of the j^z -ion. If K_a^α is taken to be 2×10^4 , 2×10^5 , and 5×10^3 for α = MIX, DFP, and DCE, respectively, the term $(RT/zF) \ln(1 + K_a^\alpha \chi c_{SE}^\alpha)$ is estimated to be 0.044, 0.070, and 0.030 V. However, the $\Delta_a^w \phi^{0'}$ values or the $\Delta G_{tr,\alpha \rightarrow W}^{0'}$ values, being defined by

$$\Delta G_{tr,\alpha \rightarrow W}^{0'} = -zF \Delta_a^w \phi^{0'}, \quad (4)$$

may be useful to discuss preliminarily the extractability of ions from W- into the α phase. The calculated $\Delta G_{tr,\alpha \rightarrow W}^{0'}$ values are listed in Table 4.

All of the ions tested gave higher $\Delta G_{tr,MIX \rightarrow W}^{0'}$ values than $\Delta G_{tr,DFP \rightarrow W}^{0'}$, indicating again that the chemical potentials of ions, including the fluorophilic ones, in MIX are lower than those in DFP. By filled circles in Fig. 3, the $\Delta G_{tr,MIX \rightarrow W}^{0'}$ values for non-fluorinated ions are plotted against $\Delta G_{tr,DCE \rightarrow W}^{0'}$. Despite the differences in the sign of the charge and in the structure among

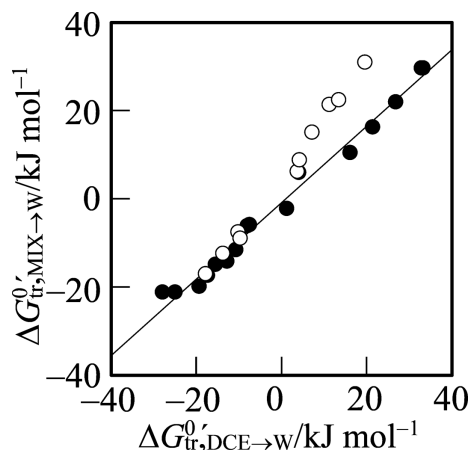


Fig. 3 Plots of $\Delta G_{tr,MIX \rightarrow W}^{0'}$ against $\Delta G_{tr,DCE \rightarrow W}^{0'}$ for non-fluorinated ions (●) and highly-fluorinated ions (○). Solid line, calculated value by Eq. (5).

the ions tested, a good linear relationship was observed between $\Delta G_{tr,MIX \rightarrow W}^{0'}$ and $\Delta G_{tr,DCE \rightarrow W}^{0'}$. The regression line is given by

$$\Delta G_{tr,MIX \rightarrow W}^{0'}/\text{kJ mol}^{-1} = (0.87 \pm 0.06)(\Delta G_{tr,DCE \rightarrow W}^{0'}/\text{kJ mol}^{-1}) + (-0.9 \pm 1.2), \quad (5)$$

with a correlation coefficient of 0.986. Therefore, the solvation of the non-fluorinated ions in MIX may be explained in the same way as that in DCE.¹⁴ In the figure, the $\Delta G_{tr,MIX \rightarrow W}^{0'}$ values for highly-fluorinated ions are plotted by open circles. The BF₄⁻ ion and C_nF_{2n+1}COO⁻ ions with $n \leq 4$ gave negative $\Delta G_{tr,DCE \rightarrow W}^{0'}$ values, so that they are categorized to hydrophilic fluorinated ions. Their $\Delta G_{tr,MIX \rightarrow W}^{0'}$ values are very close to those predicted from Eq. (5). This category of ions is not bulky, so their surfaces have a higher electric field strength. Therefore, the electrostatic ion-solvent interactions would contribute significantly to the Gibbs transfer energy.¹⁵

On the other hand, PF₆⁻, C₇F₁₅COO⁻, C₈F₁₇SO₃⁻, and (C_nF_{2n+1}SO₂)₂N⁻ are categorized to hydrophobic fluorinated ions. Their $\Delta G_{tr,MIX \rightarrow W}^{0'}$ values are higher than those predicted from Eq. (5) with increasing $\Delta G_{tr,DCE \rightarrow W}^{0'}$, indicating a higher affinity of the highly-fluorinated ions to the fluorosolvent. A similar result was observed in the $\Delta G_{tr,DFP \rightarrow W}^{0'}$ v.s. $\Delta G_{tr,DCE \rightarrow W}^{0'}$ plot.² The difference between the experimental $\Delta G_{tr,MIX \rightarrow W}^{0'}$ value for (C₃F₇SO₂)₂N⁻ as an example and the calculated one by Eq. (5) is 14.5 kJ mol⁻¹, which is only 4.3 kJ mol⁻¹ smaller than that from the $\Delta G_{tr,DFP \rightarrow W}^{0'}$ v.s. $\Delta G_{tr,DCE \rightarrow W}^{0'}$ plot. Therefore, even if the same volume of DCE is mixed, the fluorinated ions would be preferentially solvated by the fluorosolvent, DFP.

Thus, it can be stated that the addition of DCE into DFP improves the stability of ions, but maintains the fluorosolvent character. On the basis of the finding, the extraction of the C₈F₁₇SO₃⁻ ion with the α (= MIX, DFP, and DCE) | MB⁺, X⁻ (W) interfacial system was examined. After the shaking, the α phase was diluted 10 times with acetonitrile, and the absorption spectrum for the dilution was recorded. In Fig. 4, solid lines show the spectra for the dilutions prepared when the W phase contained 2 mM MB⁺ and 0.2 mM C₈F₁₇SO₃H. An absorption maximum was observed at 655 nm. When α = MIX, the absorbance, A₆₅₅, is equal to that given by a reference material (dashed line in the figure), A₆₅₅⁰, indicating that C₈F₁₇SO₃⁻ was extracted quantitatively into the MIX phase. When α = DCE, A₆₅₅ is slightly smaller than A₆₅₅⁰. When α = DFP, A₆₅₅ is

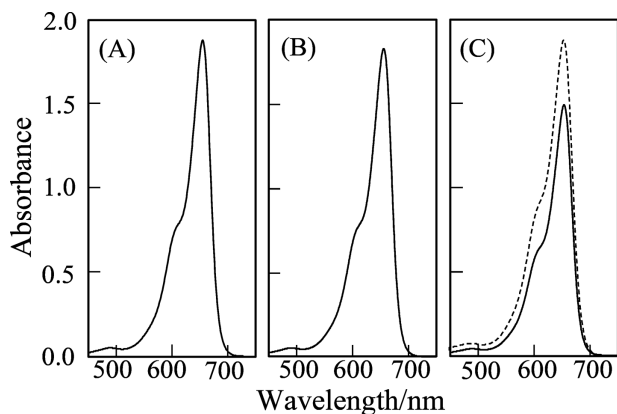


Fig. 4 Absorption spectra for the 10-fold dilutions with acetonitrile of the (A) MIX, (B) DCE, and (C) DFP phases after shaking the W phase containing 2 mM MBCl and 0.2 mM $\text{C}_8\text{F}_{17}\text{SO}_3\text{H}$. Dashed line, the 10-fold dilution of the 0.2 mM MBCl aqueous solution.

significantly smaller than A_{655}^0 . Thus, the extractability increased in order of $\text{DFP} < \text{DCE} < \text{MIX}$. The results do not contradict a simple prediction from $\Delta G_{\text{tr},\alpha \rightarrow \text{W}}^{0'}(\text{MB}^+) + \Delta G_{\text{tr},\alpha \rightarrow \text{W}}^{0'}(\text{C}_8\text{F}_{17}\text{SO}_3^-) = 25.1, 33.0, \text{ and } 37.5 \text{ kJ mol}^{-1}$ for $\alpha = \text{DFP}, \text{DCE}, \text{ and MIX}$, respectively.

In conclusion, the DFP-DCE mixed solvent is a practically superior extraction medium for fluorophilic ions. This finding would be useful in the fluorous chemistry concerned with fluorophilic ionic species.

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